

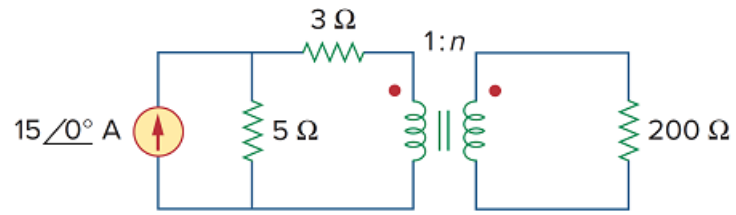
Problem

Refer to the network in Fig. 13.118.

(a) Find n for maximum power supplied to the 200- Ω load.

(b) Determine the power in the 200- Ω load if $n = 10$.

Figure 13.118



Step-by-step solution

Step 1 of 6

Refer to Figure 13.118 in the textbook for a transformer network.

(a)

Convert the current source in the circuit into voltage source.

The current in the primary side of the transformer is,

$$I_{s1} = 15\angle 0^\circ \text{ A}$$

Therefore,

$$\begin{aligned} V_{s1} &= I_{s1}R \\ &= (15\angle 0^\circ \text{ A})(5 \Omega) \\ &= 75\angle 0^\circ \text{ V} \end{aligned}$$

Add the resistances in the primary side of the circuit.

$$\begin{aligned} R_{in} &= R + 3 \Omega \\ &= 5 \Omega + 3 \Omega \\ &= 8 \Omega \end{aligned}$$

Reflect the secondary impedance of transformer to the primary side.

$$R_L = \frac{R_{L2}}{n^2}$$

Here, n is the turns ratio of the transformer. Therefore,

$$R_L = \frac{200}{n^2}$$

Step 2 of 6

Draw the simplified circuit.

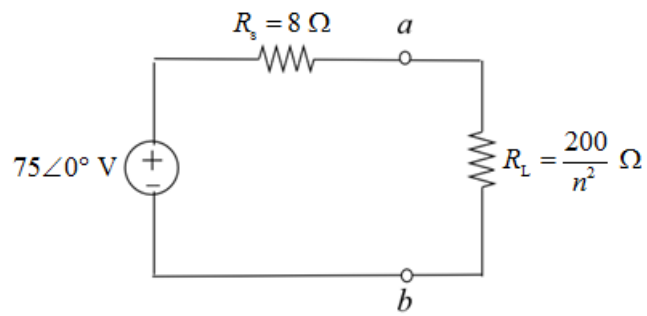


Figure 1

Step 3 of 6

Consider that the maximum power is transferred to the load when the load resistance is equal to source resistance as per the maximum power transfer theorem. Therefore,

$$R_L = R_s$$

Substitute the values of R_L and R_s .

$$\frac{200}{n^2} = 8$$

$$n^2 = \frac{200}{8}$$

$$= 25$$

$$n = 5$$

Therefore, the value of turns ratio, n for maximum power transfer in the circuit is 5.

Step 4 of 6

(b)

Consider that the turns ratio, $n = 10$.

Represent the loop currents in the circuit and draw modified circuit.

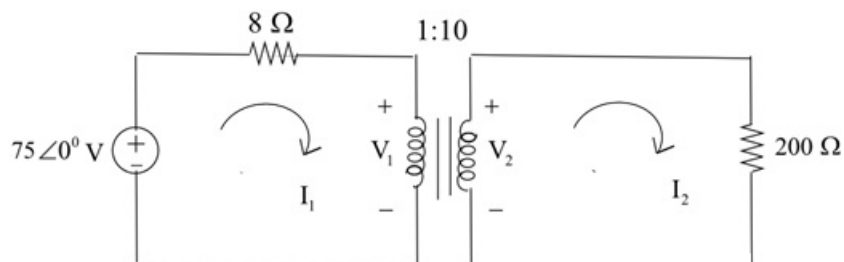


Figure 2

Step 5 of 6

Apply Kirchhoff's voltage law to input mesh in the circuit.

$$8I_1 + V_1 - 75 = 0$$

Apply Kirchhoff's voltage law to output mesh in the circuit.

$$200I_2 - V_2 = 0$$

Consider the following expression for the voltage ratio of the transformer:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

Substitute 10 for N_2 and 1 for N_1 .

$$\frac{V_2}{V_1} = \frac{10}{1}$$

$$V_1 = \frac{V_2}{10}$$

Consider the following expression for the current ratio of the transformer:

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}$$

Substitute 10 for N_2 and 1 for N_1 .

$$\frac{I_1}{I_2} = \frac{10}{1}$$

$$I_1 = 10I_2$$

Step 6 of 6

Substitute the expressions of V_1 and I_1 in the first KVL equation.

$$8I_1 + V_1 - 75 = 0$$

$$8(10I_2) + \frac{V_2}{10} - 75 = 0$$

$$80I_2 + \frac{V_2}{10} = 75$$

$$800I_2 + V_2 = 750$$

$$V_2 = 750 - 800I_2$$

Substitute the expression of V_2 in the second KVL equation.

$$200I_2 - V_2 = 0$$

$$200I_2 - (750 - 800I_2) = 0$$

$$200I_2 + 800I_2 = 750$$

$$1000I_2 = 750$$

$$I_2 = \frac{750}{1000}$$
$$= 0.75 \text{ A}$$

Calculate the power in the load resistor, 200Ω in the circuit.

$$P_L = I_2^2 R_L$$

Substitute 0.75 A for I_2 and 200Ω for R_L .

$$P_L = (0.75)^2 (200)$$
$$= (0.5625)(200)$$
$$= 112.5 \text{ W}$$

Thus, the power in the 200Ω resistor is 112.5 W.